AI-Based Car Damage Detection and Repair Cost Estimation

Research Project — B.Tech CSE AI/ML, 7th Semester

Research Question

> Can structured damage information extracted from vehicle images (component + damage type + severity + vehicle features) produce useful repair-cost estimates compared with directly predicting cost from an image?

Pipeline

```
Car Image
    ↓
Image Preprocessing
    ↓
Damage Detection (YOLO)
    ↓
Damage Localization (bounding boxes)
    ↓
Damage Type Classification
    ↓
Vehicle Component Identification
    ↓
Severity Assessment (Minor/Moderate/Severe)
    ↓
Vehicle + Cost Features
    ↓
Repair-Cost Estimation (range, not exact)
    ↓
Final Damage Report
```

Project Structure

```
car-damage-ai/
├── data/                  # Datasets
│   ├── raw/               # Original downloaded data
│   ├── processed/         # Prepared training data
│   ├── annotations/       # Converted annotations
│   └── cost_data/         # Repair cost records
├── models/                # Trained model weights
│   ├── detection/
│   ├── classification/
│   ├── severity/
│   └── cost/
├── notebooks/             # Jupyter notebooks for EDA
```

```
├── src/                    # Source code
│   ├── detection/
│   ├── classification/
│   ├── severity/
│   ├── cost/
│   ├── pipeline/
│   └── utils/
├── evaluation/             # Evaluation results
│   ├── detection/
│   ├── classification/
│   ├── severity/
│   └── cost/
├── app/                    # Web application
│   ├── backend/            # FastAPI
│   └── frontend/           # React
├── research/               # Literature review, reports
├── reports/                # Generated reports
└── experiments/            # Experiment configs and logs
```

Technology Stack

- **Language:** Python 3.10+
- **Deep Learning:** PyTorch, Ultralytics YOLO
- **CV:** OpenCV
- **Data:** NumPy, Pandas
- **ML:** Scikit-learn, XGBoost
- **Backend:** FastAPI
- **Frontend:** React + Tailwind CSS
- **Database:** SQLite (dev) → PostgreSQL (prod)
- **Deployment:** Docker

Development Phases

Phase	Description	Status
1	Research & Literature Review	✅ Complete
2	Dataset Analysis	✅ Complete
3	Data Preparation	✅ Complete
4	Baseline Damage Detector	✅ Complete
5	Damage Classification	⬜ Next
6	Component Identification	⬜
7	Severity Assessment	⬜
8	Repair-Cost Dataset	⬜
9	Cost Model	⬜

Phase	Description	Status
10	Key Research Experiment	⬜
11	Generalization	⬜
12	Error Analysis	⬜
13	Complete Pipeline	⬜
14	Web Application	⬜

Reproducibility

All experiments record: dataset version, model, hyperparameters, random seed, hardware, metrics, date.

License

Research project — not for commercial use.